Skills Summary

Diagnosing Swapped Input–Output and Notation Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — What $f(a)$ actually names

Read the notation out loud:

$f(4) = 14$ says "the function f turns the **input** 4 into the **output** 14." The number *inside* the parentheses is always the input; the number *after the equals sign* is always the output.

$f(\ )$ is the **name of a rule**, not a multiplication. So $f(2+3)$ means "apply the rule to 5" — it is not $f \cdot (2+3)$, and it is not $f(2) + f(3)$.

Example: $f(x) = 3x + 2$. Find $f(4)$.
Step 1. The 4 sits inside the parentheses, so it is the input — this is an evaluate question: substitute and compute.
Step 2. Replace every x in the rule with 4: $f(4) = 3(4) + 2 = 12 + 2$, so $f(4) = \mathbf{14}$.

Try it: $g(x) = 5x - 1$. Find $g(4)$.

check: **19**

Skill 2 — Evaluate, or solve?

The one-question test: *where is the number written?*

Given	Question type	What you do
inside the parentheses, $f(3)$	evaluate	substitute, then compute
across the equals sign, $f(x) = 3$	solve	write an equation, solve for x

The two questions run through the same rule in *opposite directions*, so they almost never have the same answer. Always check a solved input by evaluating it forwards.

Example: $f(x) = 3x + 2$. Find x when $f(x) = 17$.
Step 1. The 17 is across the equals sign from $f(x)$, so it is an output — that makes this a solve question, not a substitution.
Step 2. Write the equation and undo the rule: $3x + 2 = 17$, so $3x = 15$ and $x = \mathbf{5}$. Check forwards: $3(5) + 2 = 17$.

Try it: $g(x) = 5x - 1$. Find x when $g(x) = 34$.

check: **7**

(!) Watch out: substituting the output is the most common slip on this page. If you are given $f(x) = 17$ and you compute $f(17)$, you have answered a question nobody asked.

Skill 3 — Reading a function table both ways

Rule: a table has an **input row** (the letter in the parentheses) and an **output row** (the $f(\ )$ row). To find $h(3)$, find 3 in the *input* row and read *down*. To answer "which input gives 3?", find 3 in the *output* row and read *up*. Same table, opposite directions — so always ask which row you started in.

Example: Find $h(3)$ from this table.

t	1	2	3	4
$h(t)$	6	9	1	3

Step 1. $h(3)$ gives an input, so start in the t row — reading the $h(t)$ row first is the error this table is built to catch.
Step 2. Find 3 in the t row (fourth column of the header) and read down: $h(3) = \mathbf{1}$. Reading the other row instead would have found 3 under $t = 4$, an answer to the *reverse* question.

Try it: Find $k(2)$.

t	1	2	3	4
$k(t)$	9	5	8	2

check: **5**